

Instructions:

- to produce your answers to the questions below, you must complete code under the given template functions below
- do not change the function names nor the input arguments
- your code must return the answer; do not manually type the answer into the Jupyter notebook
- be sure to return the data type requested otherwise your work will be considered incorrect
- you may include additional code outside of these functions; it will not be graded or graded directly

```
In [1]: # Import pandas
import pandas as pd
```

```
In [2]: # Load the babynames data file into a pandas data frame called "bnDataFrame"
df = pd.read_csv('babynames.csv')
```

1. Data set range

a. What is the first year of the data set?

```
In [31]: def getFirstYr(bnDataFrame):
          first_yr = min(bnDataFrame['Year'])
          return first_yr
print(getFirstYr(df))
```

1880

b. What is the last year of the data set?

```
In [32]: def getLastYr(bnDataFrame):
          last_yr = max(bnDataFrame['Year'])
          return last_yr
print(getLastYr(df))
```

2014

2. How many unique *boy* names are there in this dataset?

```
In [33]: def numUniqBoyNames(bnDataFrame):
          filter_b = bnDataFrame['Gender'] == 'M'
          boy_rows = bnDataFrame[filter_b]
          uniq_boy_names = boy_rows['Name'].unique()
          num_uniq_boy_names = len(uniq_boy_names)
```

```

    return num_uniq_boy_names # RETURN A NUMBER (most kinds of number c
print(numUniqBoyNames(df))

```

39199

3. How many babies were named Taylor in this dataset?

```

In [34]: def numTaylors(bnDataFrame):
    filter_taylor = bnDataFrame['Name'] == 'Taylor'
    taylor_rows = bnDataFrame[filter_taylor]
    total_taylors = sum(taylor_rows['Count'])
    return total_taylors # RETURN A NUMBER (most kinds of number data
print(numTaylors(df))

```

416096

4. How many more boys were named Logan than girls in 1998?

```

In [35]: def logan98(bnDataFrame):
    filter_1 = (bnDataFrame['Name'] == 'Logan') & (bnDataFrame['Year'] == 1998)
    logan_98_rows = bnDataFrame[filter_1]
    logan_98_boys = logan_98_rows[logan_98_rows['Gender'] == 'M']['Count'].iloc[0]
    logan_98_girls = logan_98_rows[logan_98_rows['Gender'] == 'F']['Count'].iloc[0]
    return logan_98_boys - logan_98_girls # RETURN A NUMBER (most kinds of number
print(logan98(df))

```

7613

5. Top three female names in 1998

a. What were the top three female names in 1998?

```

In [36]: def top3Fs_98(bnDataFrame):
    filter_98 = bnDataFrame['Year'] == 1998
    filter_F = bnDataFrame['Gender'] == 'F'
    rows_F_1998 = bnDataFrame[filter_98 & filter_F]
    sorted_rows_F_1998 = rows_F_1998.sort_values(by='Count', ascending=False)
    top_3 = list(sorted_rows_F_1998.iloc[:3]['Name'])
    return top_3 # RETURN A LIST
print(top3Fs_98(df))

```

['Emily', 'Hannah', 'Samantha']

b. What were the counts of the top three female names in 1998?

```

In [37]: def top3F_Counts_98(bnDataFrame):

```

```

filter_98 = bnDataFrame['Year'] == 1998
filter_F = bnDataFrame['Gender'] == 'F'
rows_F_1998 = bnDataFrame[filter_98 & filter_F]
sorted_rows_F_1998 = rows_F_1998.sort_values(by='Count', ascending=False)
top_3 = list(sorted_rows_F_1998.iloc[:3]['Count'])
return top_3 # RETURN A LIST
print(top3F_Counts_98(df))

```

[26179, 21371, 20191]

6. Unisex names

a. What are the unisex names in this dataset?

```

In [40]: def unisexNames(bnDataFrame):
    b_names = bnDataFrame[bnDataFrame['Gender']=='M']['Name'].unique()
    g_names = bnDataFrame[bnDataFrame['Gender']=='F']['Name'].unique()
    uni_names = set(g_names).intersection(set(b_names))
    return uni_names # RETURN A SET
uni_names = unisexNames(bnDataFrame)

```

```

In [41]: uni_names_list = sorted(list(uni_names))
print(uni_names_list[:5], uni_names_list[-5:])

['Aaden', 'Aadi', 'Aadyn', 'Aalijah', 'Aaliyah'] ['Zyian', 'Zyien', 'Zyion', 'Zyon', 'Zyree']

```

b. How many unisex names are in this dataset?

```

In [39]: def numUnisexNames(bnDataFrame): # HINT: reuse most of your code from t
    b_names = bnDataFrame[bnDataFrame['Gender']=='M']['Name'].unique()
    g_names = bnDataFrame[bnDataFrame['Gender']=='F']['Name'].unique()
    uni_names = set(g_names).intersection(set(b_names))
    return len(uni_names) # RETURN A NUMBER (most kinds of number data types)
numUnisexNames(bnDataFrame)

```

Out[39]: 10221

```

In [4]: from numbers import Number
funcList = ['getFirstYr', 'getLastYr', 'numUniqBoyNames',
            'numTaylors', 'logan98', 'top3Fs_98', 'top3F_Counts_98', 'unisexNames', 'numUnis
typesList = [Number, Number, Number, Number, Number, Number, list, list, set, Number]

```

Below is an optional checker you might find helpful in making sure your functions are outputting the correct data type

```

In [ ]: for i, func in enumerate(funcList):
    print(str(i) + '. ' + func)
    try:
        stuAns = eval(func + '(bnDataFrame)')
        print('Is student answer the correct data type?')
        print(isinstance(stuAns, typesList[i]))
    
```

```
except:  
    print('Student program did not work.\n')
```

7. Write code to find something interesting about your name in the dataset. Then write what you found.

There is no code template for you to use on this question. Write whatever code you need and then you can answer the question by outputting a written answer from your code or else typing your response in a subsequent Markdown cell.

Answers will vary